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Volatility Spillovers, Causality and Sensitivity: An Empirical Analysis of Ethena USDe compared to Fiat-Backed Alternative Stablecoin USDC.

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Agenda

1. Motivation and introduction of stablecoins
2. Research question and hypotheses
3. Methodology
4. Data and conditions
5. Volatility and correlation modeling
6. Main hypothesis testing methods (Granger, IRF, CoVaR)
7. Summary and implications
8. Conclusion, limitations and future research
9. (If time permitting robustness checks)

Motivation

- Learn, understand and apply time series volatility analysis, methods and processes on stablecoin and crypto assets
- Test hypotheses on stablecoin peg design and crypto-stablecoin systems and its risk dynamics
- Apply a multi-layer methodological framework of systematic volatility analysis

USDe vs USDC protocols

USDe

\$4.095B*

USDC

\$76.717B*

Backed by
crypto assets
+ short
perpetual
futures

Peg stability
comes from
delta-neutral
hedging

Fully backed
by cash +
short-term
U.S.
Treasuries

Always
redeemable
1:1 for USD

Peg stability
comes from
traditional
financial
collateral

Minimal
exposure to
crypto
market
volatility

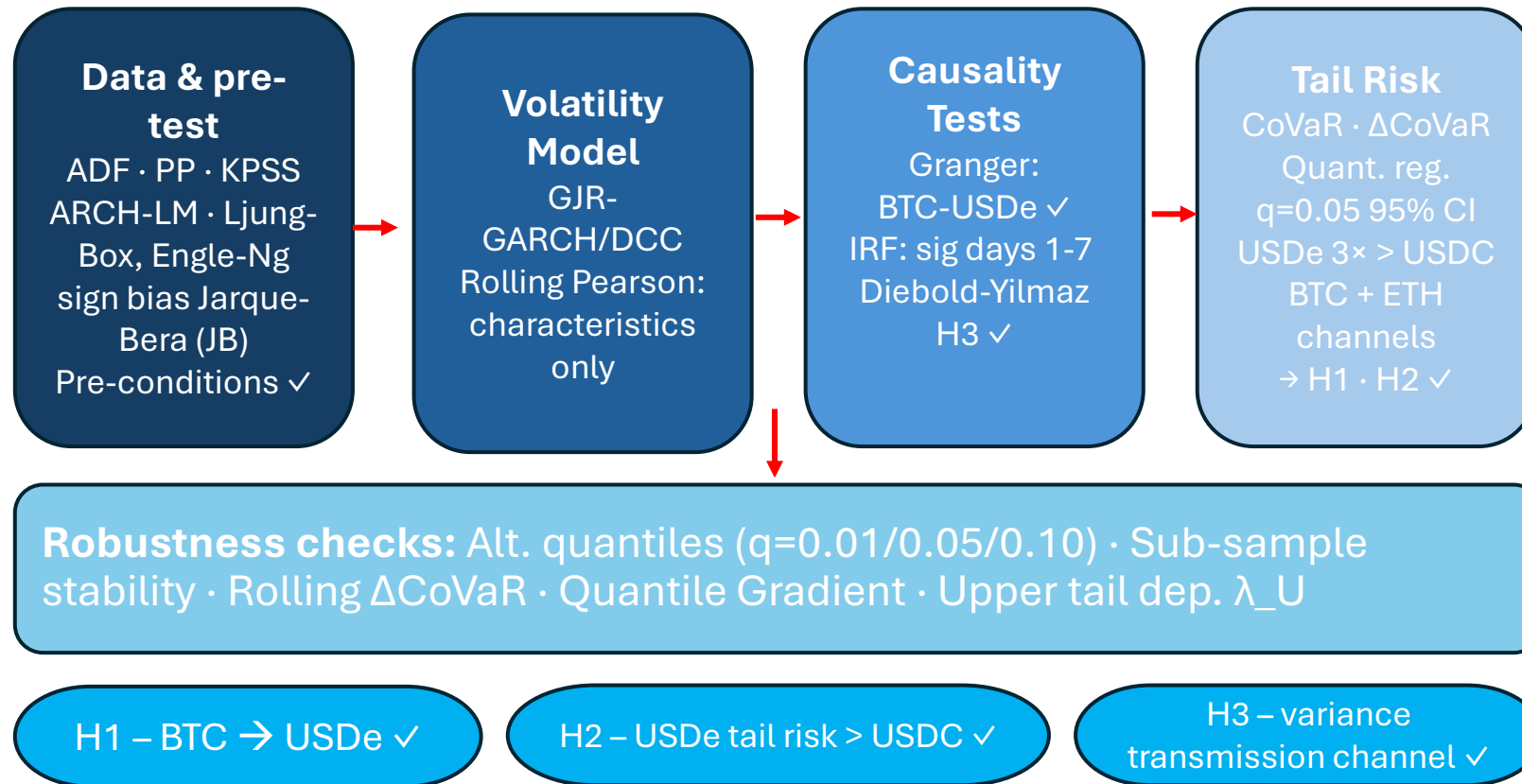
Source: Yahoo Finance, Market Cap as of 5/14/2026*

Research Question and Hypotheses

To what extent does volatility of major crypto-assets like BTC and ETH transmits into synthetic stable-coin Ethena which uses Delta-hedging and is this effect significantly different from fiat-backed stable-coin USDC?

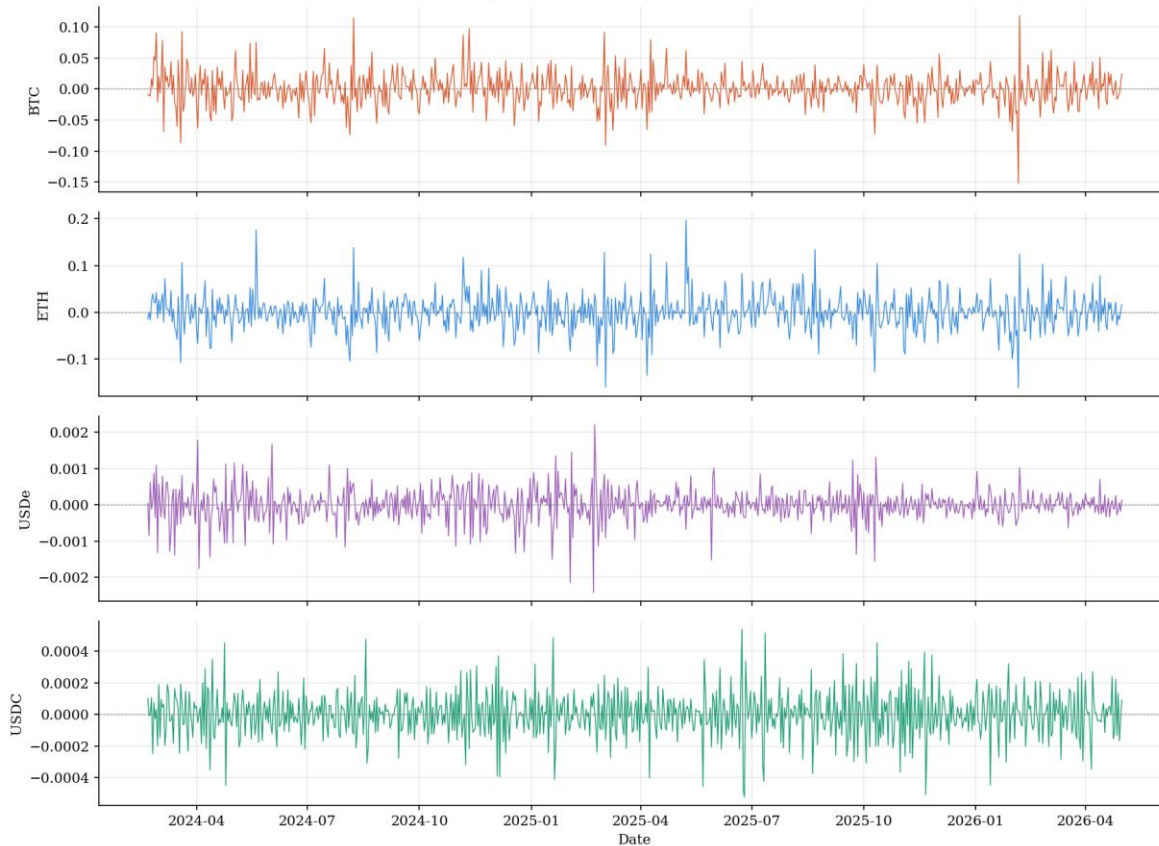
- **H1 - Volatility Spillover:** BTC, ETH volatility has a statistically significant positive effect on stablecoin peg deviation.
- **H2 - Differential Sensitivity:** The sensitivity of USDe to crypto volatility is significantly different than that of USDC.
- **H3 - Channel Identification:** Crypto risk transmission into USDe operates through the volatility channel, not through return-level dependencies.

Methodology



Data and Descriptive Statistics

Figure 1 — Daily log returns: BTC, ETH, USDe, USDC



Source: Yahoo Finance (USDE29470-USD, USDC-USD, BTC-USD, ETH-USD)

Observations: 801

Sample timeframe: 2024-02-21 → 2026-05-01

Asset	Mean	Std Dev	Skewness	Kurtosis
BTC	0.0005	0.0257	0.1294	6.5100
ETH	-0.0003	0.0372	0.2443	6.4083
USDe	-0.0000	0.0004	-0.3201	6.8087
USDC	-0.0000	0.0001	-0.0349	4.1872

All series reject normality using Jarque-Bera normality test which justifies the Student-t innovation distribution.

Non-stationarity rejection and ARCH effects

- **ADF, PP H0:** unit root (non-stationary) → reject at $p < 0.05$ = stationary evidence
- **KPSS H0:** stationary → fail to reject at $p > 0.05$ = stationary evidence

Asset	ADF p-value	PP p-value	KPSS p-value	Stationary
BTC	0.0000	0.0000	≥ 0.10	Yes ***
ETH	0.0000	0.0000	≥ 0.10	Yes ***
USDe	0.0000	0.0000	≥ 0.10	Yes ***
USDC	0.0000	0.0000	≥ 0.10	Yes ***

ARCH-LM H0: no ARCH effects (constant variance)

Mean sq residual	Ljung-Box lag 10 p-value	ARCH p-value	GARCH Justified
0.0006	0.0000***	0.0000 ***	Yes
0.0014	0.0082***	0.0197 **	Yes
0.0000	0.0000***	0.0000 ***	Yes
0.0000	0.0495*	0.0563	Borderline

Stage 1 GJR -GARCH model

- α : sensitivity of conditional variance to new shocks
- γ : extra volatility from negative shocks
- β : persistence of conditional variance
- ν : heaviness of tails / extreme co-movements
- $\alpha + \beta + \gamma/2$: overall memory of the conditional variance process

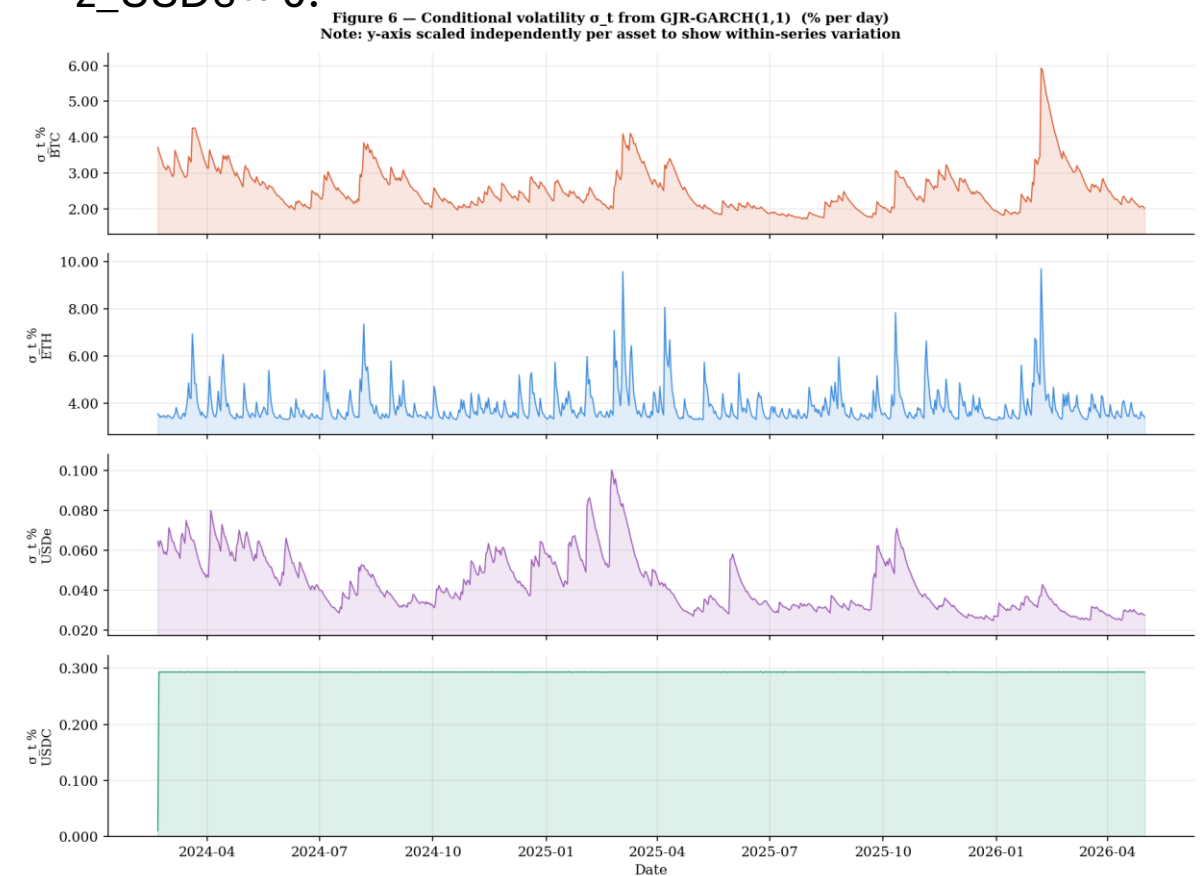
Asset	α	γ	β	ν	$\alpha + \beta + \gamma/2$ (persistence)	Half-Life (in days)
BTC	0.071	0.01	0.92	3.1	0.99	8.62
ETH	0.091	0.04	0.87	3.4	0.986	5.16
USDe	-	-	0.90	5.98	0.980	6.94
USDC	-	-	~0.001	14.16	0.595	<1d

GJR-GARCH Model Visual Evidence

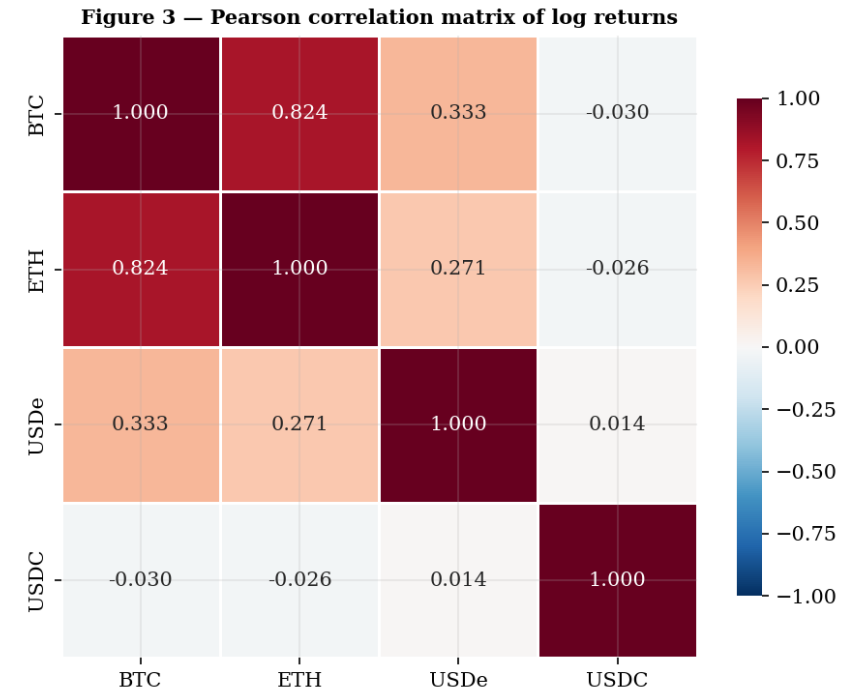
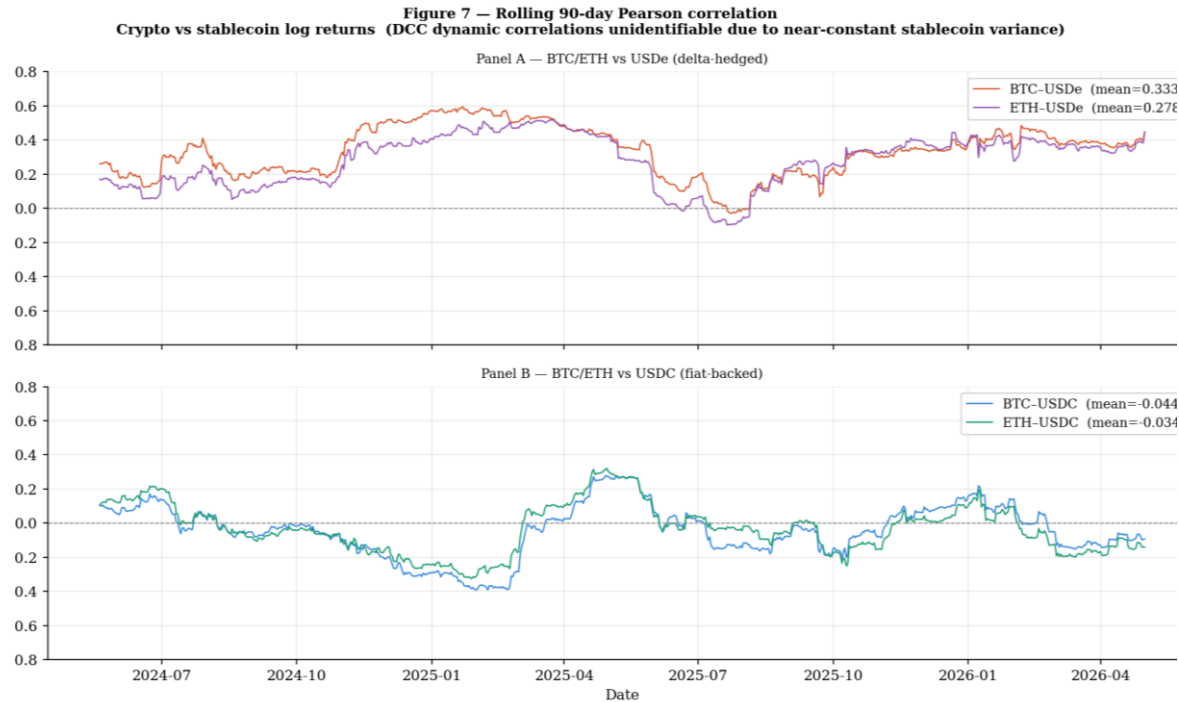
Periods of calm vs periods of spikes.

Spike = (unusual) volatility burst

DCC unidentifiable for USDC stablecoin pair. Near constant variance makes the cross-product $z_{BTC} \times z_{USDe} \approx 0$.



Dynamic Pearson Correlation



Model-free DCC alternative, 90-day rolling Pearson correlations to represent time-varying co-movement between crypto and stablecoin returns.

Granger Causality and Transmission channel

Log returns test: H0: past BTC/ETH returns have no predictive power over future stablecoin returns (testing mean channel)

Squared returns test: H0: past BTC/ETH variance (r^2) has no predictive power over future stablecoin variance (testing volatility channel).

Test on Log returns

Direction	F statistic	P-value	Significant
BTC → USDe	2.6910	0.0680	No
ETH → USDe	1.1384	0.3205	No
BTC → USDC	1.1347	0.3217	No
ETH → USDC	1.7783	0.1691	No

Test on Squared returns

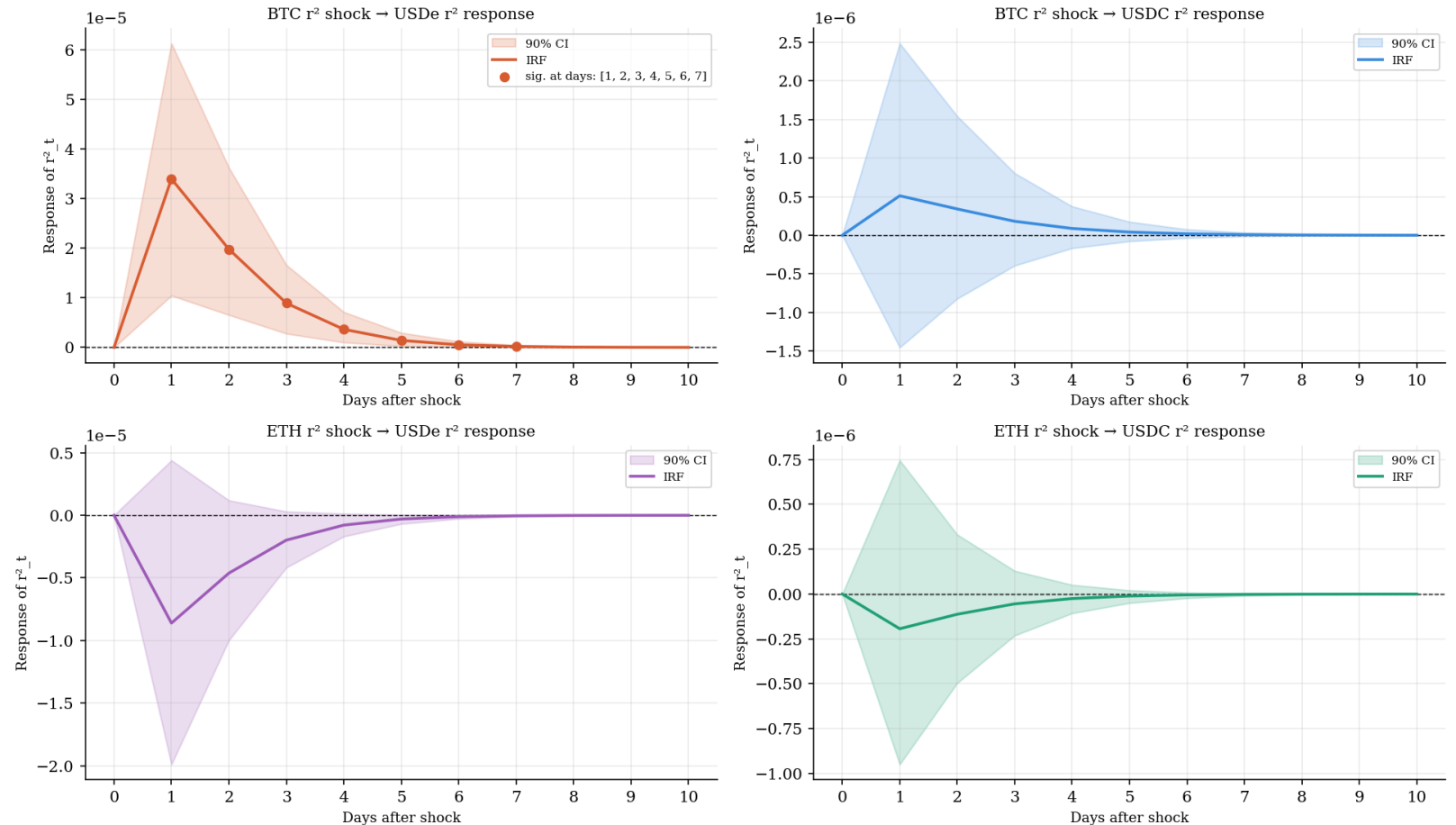
Direction	F statistic	P-value	Significant
BTC → USDe	5.4444	0.0197 **	Yes
ETH → USDe	1.4995	0.2208	No
BTC → USDC	0.2072	0.6490	No
ETH → USDC	0.1270	0.7216	No

Impulse Response Function

BTC → USDe: significant transmission confirmed

A one standard deviation shock to BTC squared returns raises USDe squared returns significantly for approximately one week before dissipating.

Figure 13a — IRF: crypto variance shock → stablecoin variance response
VAR(1) on squared returns r^2_t | Horizon: 10 days | Shaded = 90% percentile bootstrap CI | Dots = horizons where CI excludes zero



Diebold-Yilmaz Spillover Index

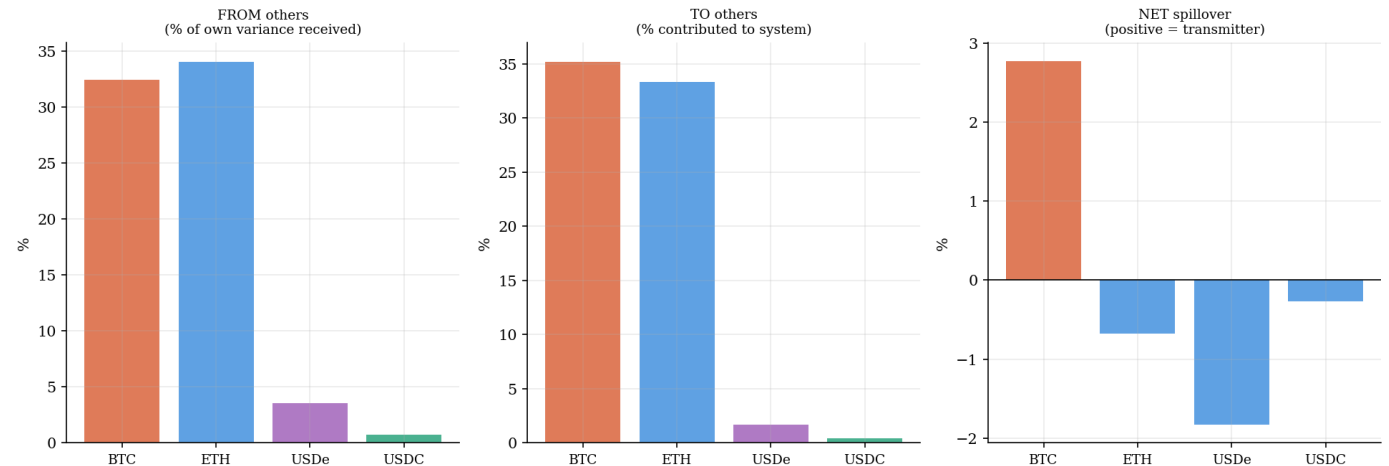
- D-Y spillover index provides a descriptive quantification of directional variance transmission
- BTC shocks account for 2.16% of USDe's 10-step forecast error variance compared to 0.06% for USDC

Forecast error of [row] explained by shocks from [col]

To/From	BTC	ETH	USDe	USDC
BTC	67.59	31.64	0.75	0.02
ETH	32.97	65.98	0.75	0.31
USDe	2.16	1.24	96.48	0.12
USDC	0.06	0.46	0.19	99.29

Total Spillover Index: 17.67%

Figure 8 — Diebold-Yilmaz directional volatility spillovers
Input: squared log returns r^2_t | Forecast horizon $H=10$



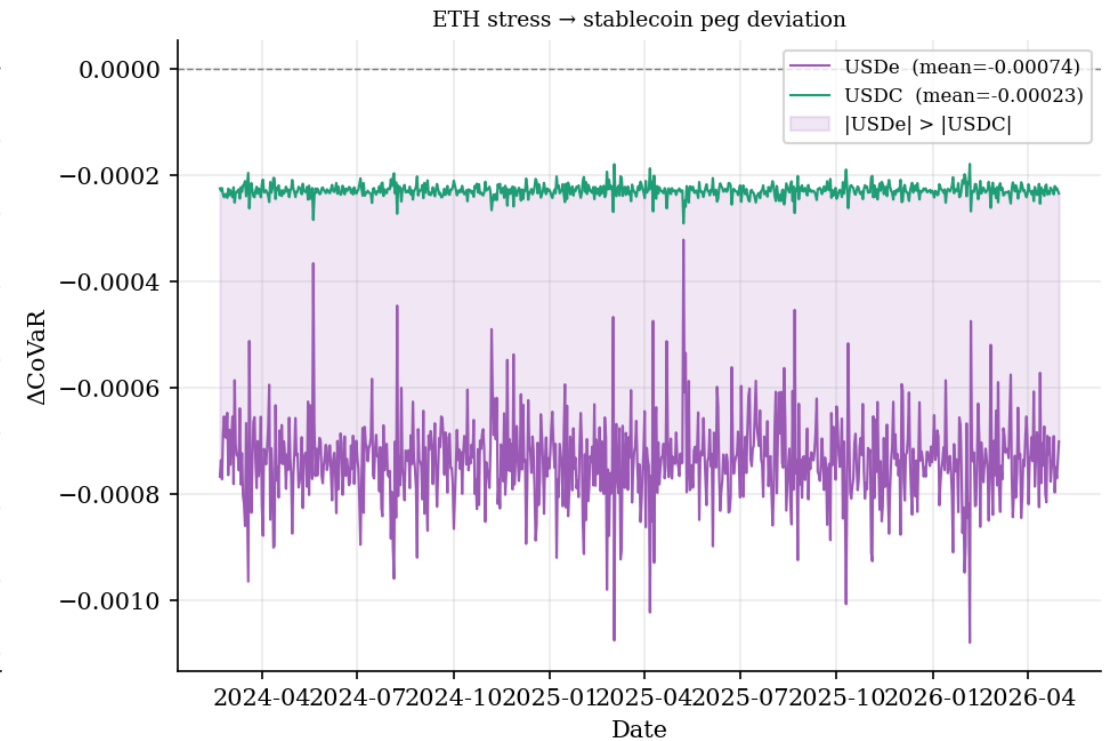
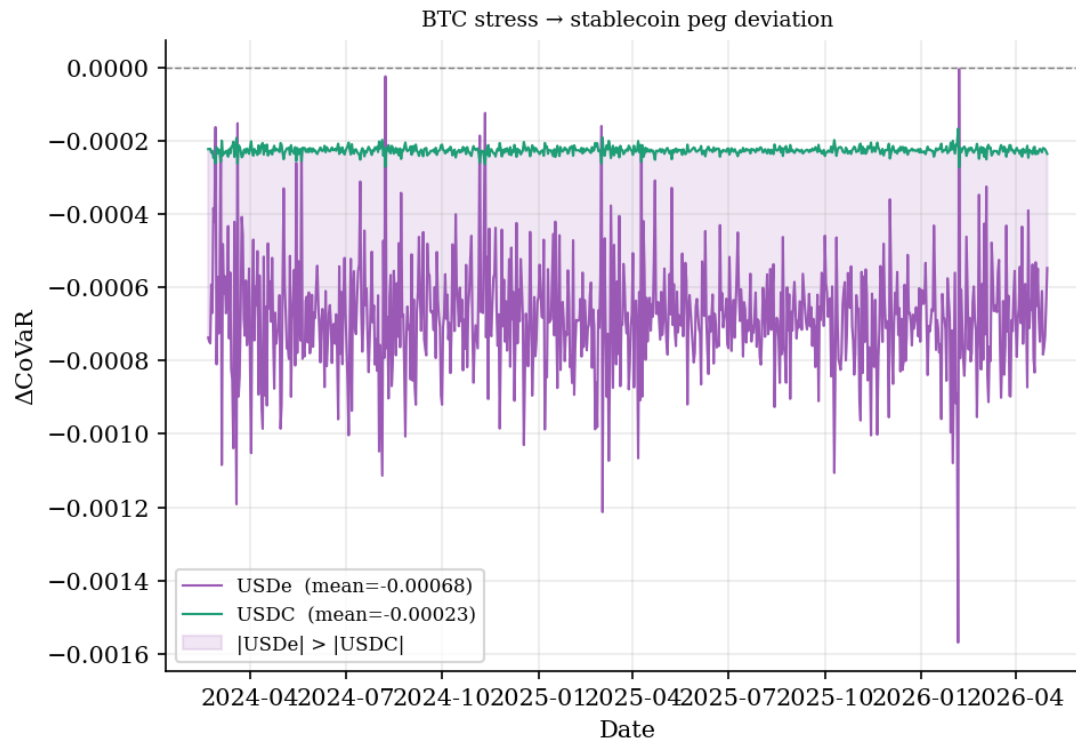
CoVaR – Tail Risk Transmission

- CoVaR table at 0.05 quantile (distress) vs 0.5 quantile (normal)
- USDe Absorbs 3× More Tail Risk Than USDC During Crypto Stress
- H2 Confirmed ($|USDe \text{ Mean } \Delta CoVaR| > |USDC \text{ Mean } \Delta CoVaR|$)

Pair	$\beta(q=0.05)$	P-value	$\beta(q=0.50)$	P-value	Mean $\Delta CoVaR$	CI Exclude 0	Sig.	$ USDe > USDC $
BTC-USDe	0.011306	0.000	0.005517	0.000	-0.000685	TRUE	YES	TRUE
BTC-USDC	-0.000382	0.413	0.000006	0.980	-0.000226	TRUE	NO	
ETH-USDe	0.005193	0.009	0.003083	0.000	-0.000738	TRUE	YES	TRUE
ETH-USDC	-0.000331	0.319	-0.000019	0.904	-0.000229	TRUE	NO	

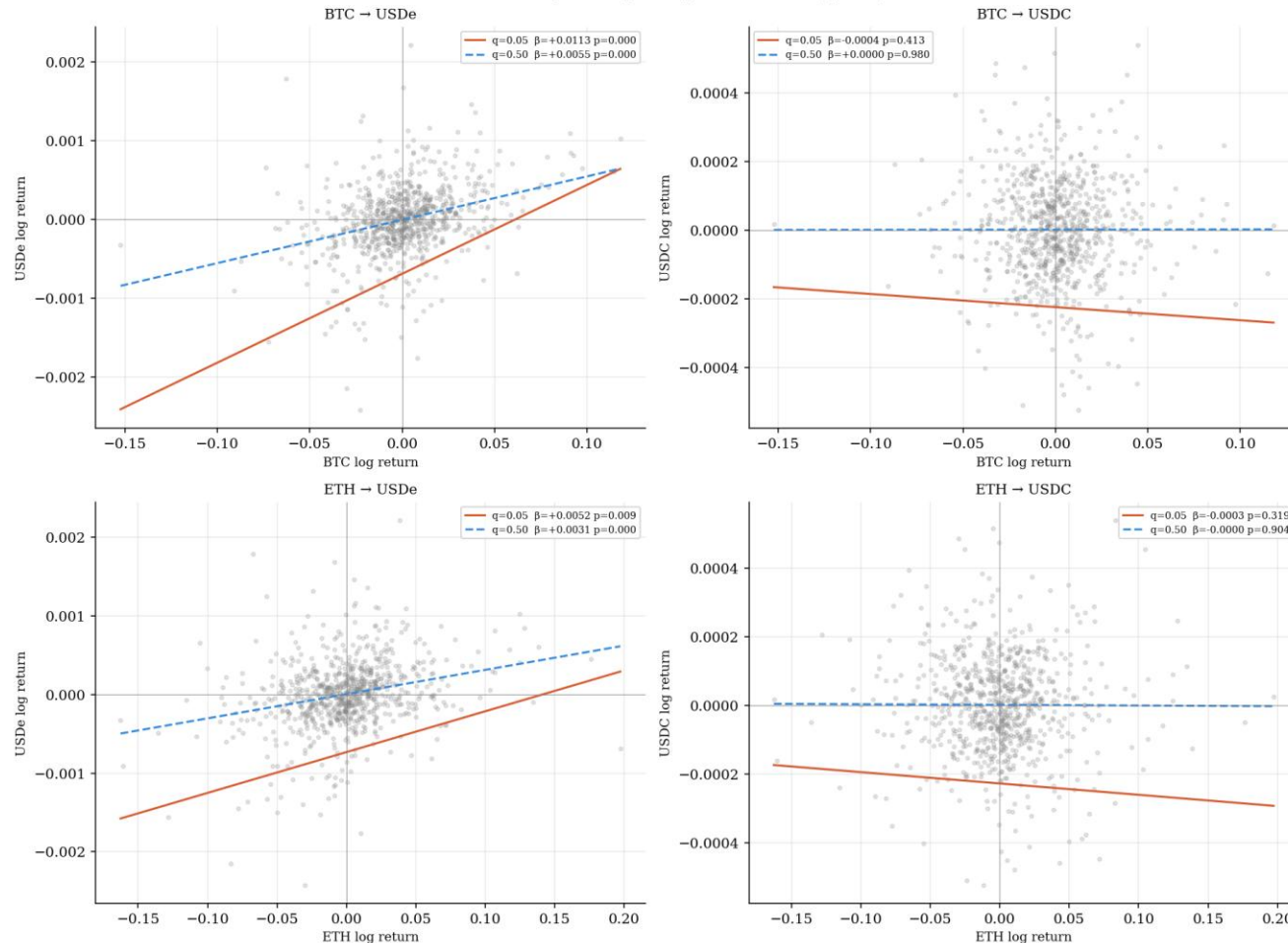
CoVaR – Tail Risk Transmission

Figure 9 – ΔCoVaR time series: USDe vs USDC
 $\Delta\text{CoVaR} = \text{CoVaR}(\text{distress } q=0.05) - \text{CoVaR}(\text{median } q=0.50)$



CoVaR quantile regression and correlations

Figure 10 — Quantile regression: crypto stress vs stablecoin returns
Red = distress quantile ($q=0.05$), Blue = median ($q=0.50$)



Does the *slope* of the relationship between crypto volatility and SC returns differ from zero?

If $p>0.05$: cannot distinguish that slope from a flat line. Crypto stress has no measurable marginal effect on SC tail outcome.

A 1% increase in BTC conditional volatility moves USDC's 5th percentile by 0.000382 ppts vs 0.011 ppts. of USDe.

USDe steeper slope at $q=0.05$ vs $q=0.50$
→ Direction consistent with tail strengthening.

Summary of Findings

- H1 Volatility Spillover: BTC volatility transmits to USDe
- H2 Sensitivity: USDe is more sensitive than USDC to crypto tail risk
- H3 Confirmation of volatility as transmission channel between crypto assets and USDe vs USDC via Granger on squared returns.

Hypothesis	Method	BTC → USDe	ETH → USDe	BTC → USDC	ETH → USDC
H1	Granger, IRF	Significant	Not sig.	Not sig	Not sig.
H2	ΔCoVaR	Significant	Significant	Not sig.	Not sig.
H3	Granger	Significant	None	None	None

Implication / Interpretation

Portfolio Implication

USDe adds tail risk exactly when crypto markets are stressed meaning it behaves *less* like a safe asset than its peg implies.

Market Implication

BTC volatility and USDe peg instability move together in stress tail, revealing the limits of delta-hedging under stress.

Regulatory Implication

Synthetic and reserve-backed stablecoins behave differently in tail events, supporting distinct regulatory contemplation.

Conclusions & limitations

Conclusion Statements

1. **Based on results given our data set and methodology, crypto volatility transmits into synthetic stablecoin peg dynamics, but not into reserve-backed stablecoins.**
2. **Volatility spillovers are present but modest at the system level (~17%).**
3. **Stablecoin peg designs show different tail risk dynamics between fiat-backed and synthetic.**

Limitations

1. **Short sample period**
2. **Comparing to one fiat-backed stablecoin**
3. **Daily data missed intraday information and stress transmission**
4. **DCC-GARCH unidentified for crypto-stablecoin pairs**
5. **Granger causality measures predictability, not structural causation**

Future Research

- 1. Hawkes/Poisson peg-deviation event models**
- 2. Include Implied Volatility estimated through crypto futures open interest**
- 3. Include more fiat-backed SC into the analysis**
- 4. ML models to analyze volatility clusters**
- 5. Perform methods with high frequency intraday data**
- 6. As USDe further grows, conduct longer timeframe analysis.**



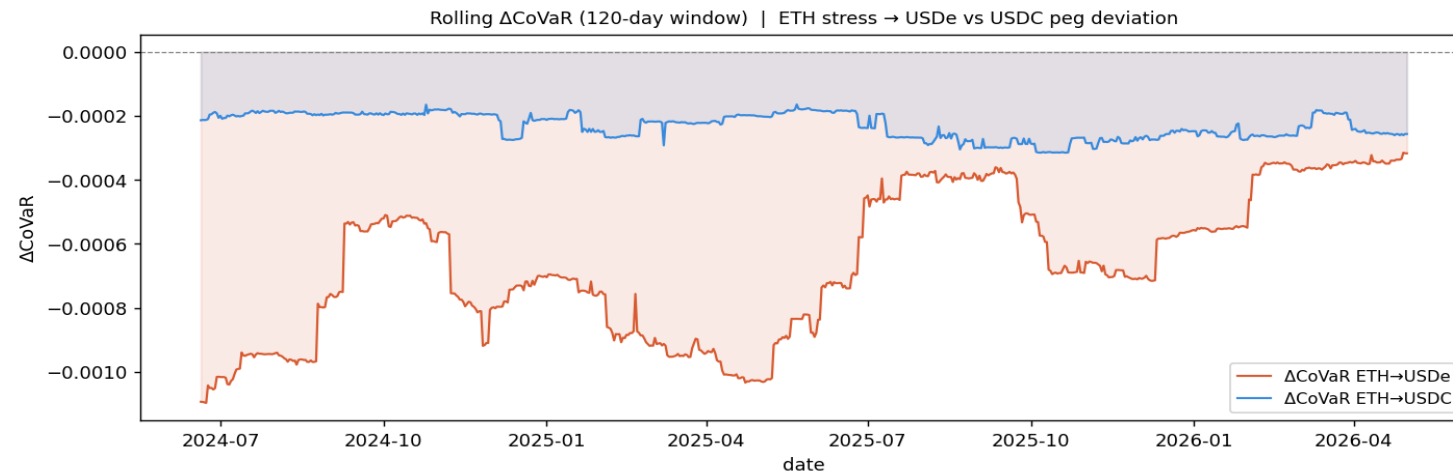
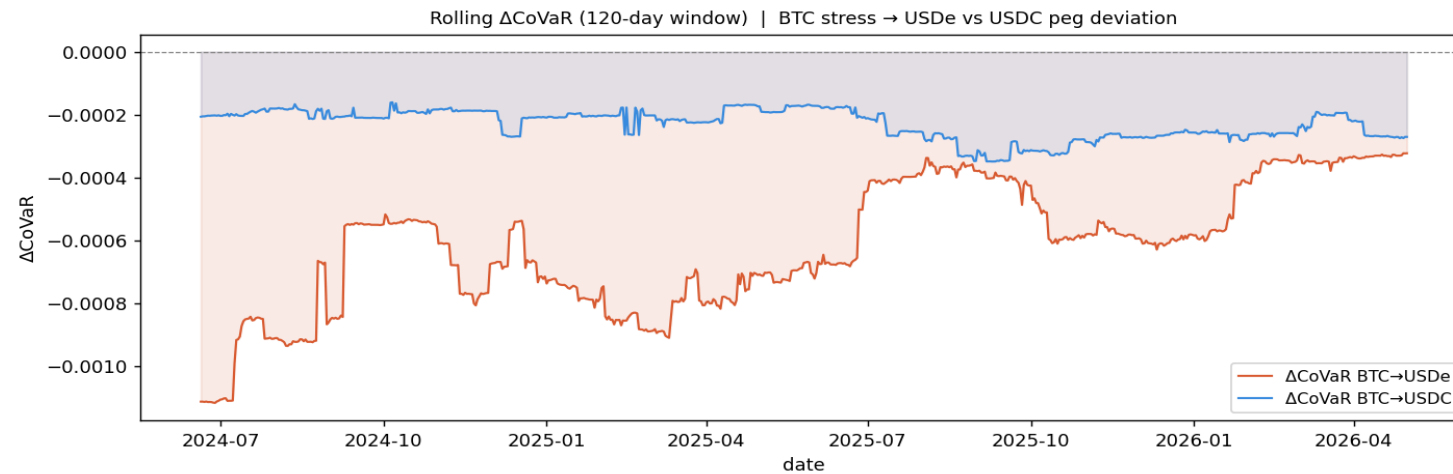
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Thank you.

Robustness Check Rolling ΔCoVaR



Robustness Check Alt. quantiles & periods

Quantile	Crypto	USDe	USDC	Holds
0.01	BTC	0.001212	0.000400	TRUE
0.01	ETH	0.001244	0.000402	TRUE
0.05	BTC	0.000685	0.000226	TRUE
0.05	ETH	0.000738	0.000229	TRUE
0.1	BTC	0.000430	0.000174	TRUE
0.1	ETH	0.000447	0.000174	TRUE

Check:

$|\Delta\text{CoVaR}(\text{USDe})| > |\Delta\text{CoVaR}(\text{USDC})|$ at different quantiles

Pair	Period 1	Period 2	Holds
BTC-USDC	-0.000213	-0.000264	TRUE
BTC-USDe	-0.000847	-0.000418	TRUE
ETH-USDC	-0.000216	-0.000261	TRUE
ETH-USDe	-0.000940	-0.000414	TRUE

Check:

Period 1 (2024-02-21 – 2025-03-27)
N=400

Period 2 (2025-03-27 – 2026-05-01)
N=401

Values = mean ΔCoVaR at $q=0.05$

Robustness Check QGB

Figure A2 — Quantile Gradient Boosting: BTC and ETH feature importance across quantiles
Left: USDe target | Right: USDC target
Rising importance at upper quantiles = transmission concentrates in high-variance episodes

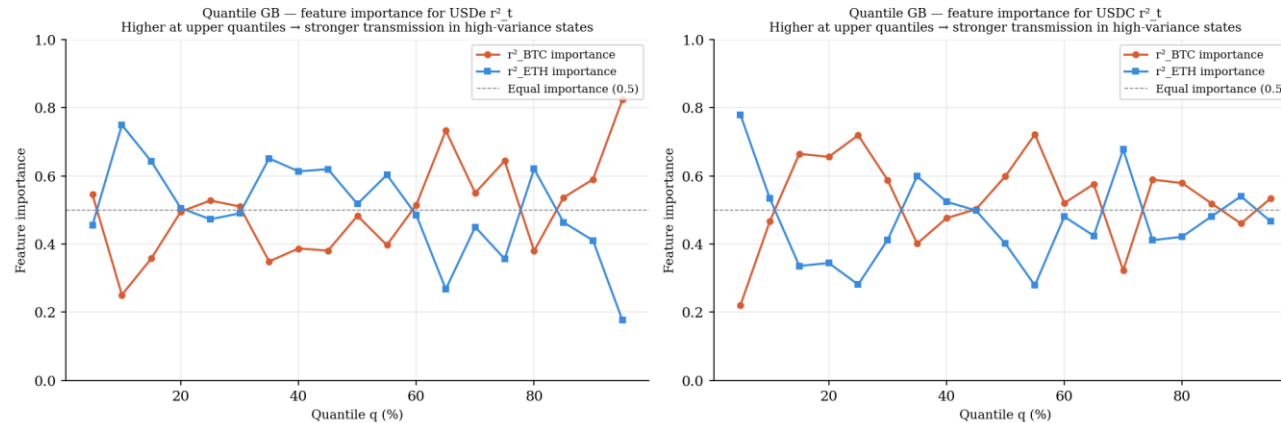
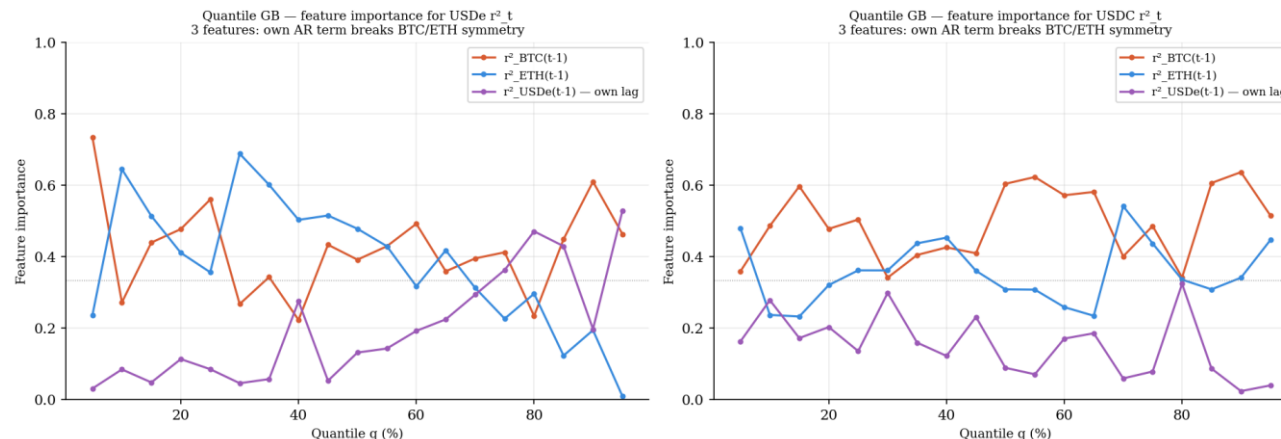


Figure A2 (corrected) — Quantile Gradient Boosting feature importance
Left: USDe target | Right: USDC target | Features: $r^2_{BTC}(t-1)$, $r^2_{ETH}(t-1)$, $r^2_{USDe}(t-1)$

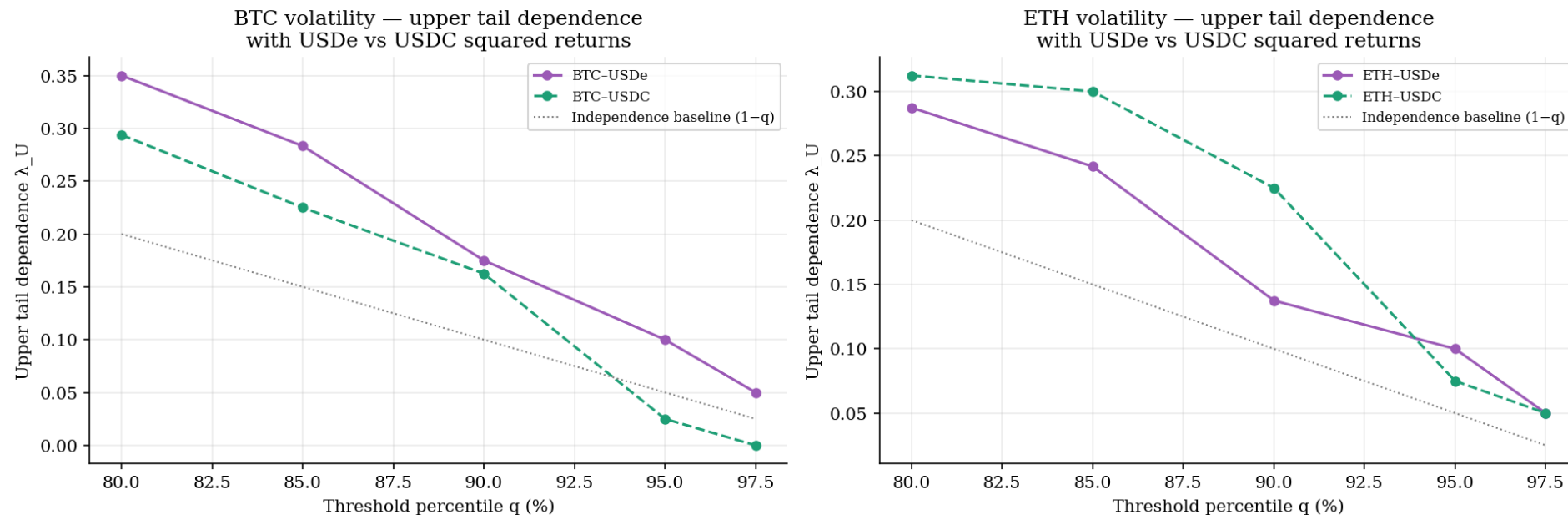


- A model-free machine learning approach
- no linearity or distributional assumptions
- estimates which crypto predictor drives stablecoin variance at each quantile of the distribution.
- Rising BTC importance at upper quantiles ($q > 80\%$) confirms that the BTC→USDe transmission concentrates in high-variance states, consistent with Granger and CoVaR findings.

Robustness Check upper tail dependence

Non-parametric empirical upper tail dependence. The coefficient λ_U measures how often BTC and USDe are simultaneously in their most extreme variance days, compared to the 10% expected by pure chance at the 90th percentile threshold. No model and distribution assumption.

Figure A1 — Empirical upper tail dependence λ_U across thresholds
Values above the independence baseline indicate joint extreme co-movement.
Higher λ_U for USDe than USDC confirms H2 nonparametrically.



Appendix

Three sub-tests on GARCH(1,1) standardized residuals:

H0 for each: no asymmetric effect (symmetric GARCH is sufficient)

Rejection → asymmetric GARCH (GJR or EGARCH) is formally justified.

- Sign Bias: do negative vs positive shocks affect future volatility differently?
- Negative Size: do LARGE negative shocks have extra impact beyond sign alone?
- Positive Size: do LARGE positive shocks have extra impact beyond sign alone?
- Joint test: F-test of all three simultaneously (the key result)

```
Engle-Ng Sign Bias Test on GARCH(1,1)-t residuals
H0: no asymmetric volatility effect (symmetric GARCH sufficient)
Rejection → GJR-GARCH gamma term is statistically justified
=====

BTC
  Sign Bias:      t = +0.4077  p = 0.6836
  Negative Size:  t = -2.4897  p = 0.0130
  Positive Size:  t = +0.9377  p = 0.3487
  Joint F-test:   F = 6.6628  p = 0.0002*** → GJR justified ✓

ETH
  Sign Bias:      t = +3.0928  p = 0.0021
  Negative Size:  t = +0.9764  p = 0.3292
  Positive Size:  t = +0.0421  p = 0.9664
  Joint F-test:   F = 9.0563  p = 0.0000*** → GJR justified ✓

USDe
  Sign Bias:      t = -1.7690  p = 0.0773
  Negative Size:  t = -1.6339  p = 0.1027
  Positive Size:  t = +0.9274  p = 0.3540
  Joint F-test:   F = 2.8343  p = 0.0373** → GJR justified ✓

USDC
  Sign Bias:      t = +1.0008  p = 0.3172
  Negative Size:  t = -0.6579  p = 0.5108
  Positive Size:  t = +3.4456  p = 0.0006
  Joint F-test:   F = 12.9706  p = 0.0000*** → GJR justified ✓
```

Appendix

- GARCH model selection tables

Table 5a – GARCH Model Selection: AIC

✓ = preferred specification per asset (lowest AIC)

spec	EGARCH(1,1)-t	GARCH(1,1)-t	GJR-GARCH(1,1)-t
asset			
BTC	3659.11	3658.17	3650.18 ✓
ETH	4255.82	4254.63	4249.22 ✓
USDC	-4596.24 ✓	-4580.3	70297.8
USDe	-2919.82 ✓	-2916.16	-2911.87

Table 5a – GARCH Model Selection: BIC

spec	EGARCH(1,1)-t	GARCH(1,1)-t	GJR-GARCH(1,1)-t
asset			
BTC	3682.54	3681.6	3678.30 ✓
ETH	4279.25	4278.06	4277.33 ✓
USDC	-4572.82 ✓	-4556.87	70325.92
USDe	-2896.39 ✓	-2892.73	-2883.76

Appendix

- VAR Lag Length Selection

Table 7 – VAR Lag Length Selection
Optimal lag: AIC = 8 | BIC = 2 | Selected: 2 (BIC)

	AIC	BIC
lag		
1	-48.470500	-48.353400
2	-48.625500	-48.414500
3	-48.647100	-48.342000
4	-48.647800	-48.248500
5	-48.672700	-48.178900
6	-48.674000	-48.085500
7	-48.673300	-47.990000
8	-48.690100	-47.911800
9	-48.685900	-47.812300
10	-48.663000	-47.694100

Table 11 – CoVaR and Δ CoVaR Results
Bootstrap 95% CI on mean Δ CoVaR (1000 replications)
Sig CI = True if bootstrap CI excludes zero

	$\beta(q=0.05)$		$\beta(q=0.50)$		Mean Δ CoVaR	CI lo	CI hi	Sig CI
Pair								
BTC→USDe	0.011306	***	0.005517	***	-0.000685	-0.000807	-0.000573	True
BTC→USDC	-0.000382		0.000006		-0.000226	-0.000268	-0.000207	True
ETH→USDe	0.005193	***	0.003083	***	-0.000738	-0.000871	-0.000606	True
ETH→USDC	-0.000331		-0.000019		-0.000229	-0.000269	-0.000208	True